

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

LABORATORY INVESTIGATION & FORENSIC EVIDENCE REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Lab Title: SBT-DF203 Lab 2: HTTP Analysis Using Wireshark – Embedded Image Traffic

Course Name: Basic Networking Skills for Digital Forensics

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Operating Platform: Kali Linux 2026 (Rolling Release)

Date of Submission: 08/09/2026

Traffic Scope: Loopback (lo) Multi-Object Web Traffic

Report Version: Version 1.0 (Final Release)

Executive Summary

This report documents the forensic analysis of a plaintext HTTP session containing an embedded image. A local Apache webpage (image.html) was created with an embedded JPEG (lab_photo.jpg). Using TShark, the browser-generated traffic was captured on the loopback interface. Analysis revealed that the browser made separate GET requests for the HTML and the image. The image response was split across multiple TCP segments and reassembled by Wireshark. The image was exported using the --export-objects feature of TShark. The SHA-256 hash of the exported image matched the original source image exactly, proving byte-for-byte recovery. A curl-only comparison was also performed to demonstrate the difference between browser and curl behavior regarding embedded resources.

1. Authority, Scope, and Evidence Description

- **Authority:** This lab was performed in an isolated Kali Linux environment. No third-party systems were targeted.
- **Scope:** Analysis of a single HTTP session for image.html and lab_photo.jpg.
- **Evidence Files:** evidence/image_traffic.pcapng (Browser capture) and evidence/curl_only.pcapng (Curl capture).
- **Source Objects:** /var/www/html/image.html and /var/www/html/lab_photo.jpg.

2. Environment and Acquisition Method

- **Web Server:** Apache2 (Active, listening on TCP port 80).
- **Capture Tool:** TShark on lo (Loopback), filter: tcp port 80.
- **Browser:** Firefox (Private/Incognito Mode).
- **Client Tool (Part F):** cURL 8.21.0.

3. Part C: Proving Two Objects Were Requested

The capture revealed two distinct primary HTTP requests initiated by the browser, alongside an automated request for the site favicon:

Frame	Time	Method	URI	Host
4	10.605664	GET	/image.html	127.0.0.1

8	10.837300	GET	/lab_photo.jpg	127.0.0.1
11	10.866501	GET	/favicon.ico	127.0.0.1

- **Browser-Generated Favicon Probe:** Note: The favicon.ico request is browser-generated. It resulted in a 404 Not Found, which is expected as no favicon was created.

4. Part D: TCP Segmentation and Reassembly

The image response (HTTP/1.1 200 OK, Content-Type: image/jpeg, Content-Length: 13232) was split across multiple TCP segments:

Frame	TCP Segment Length (tcp.len)	Sequence Number
9	13520	492
11	458	927
12	527	14012

- **Total TCP Payload:** $13520 + 458 + 527 = 14505$ bytes.
- **HTTP Header Overhead:** $14505 - 13232 = 1273$ bytes.
- **Encapsulation Formula Verified:** This confirms the encapsulation relationship: TCP Payload = HTTP Headers + HTTP Object Data.

5. Part E: Exporting the Image and Verifying Integrity

The image was exported using `tshark --export-objects http,exported/http_objects`.

Artifact Target	Cryptographic SHA-256 Hash Digest	Verification
Original (/var/www/...)	24b8a153d8d4a0f4f0e65ea5e12f3a9a5d21dd8f8e8200126e4b6a314b7d4	SOURCE
Exported (exported/...)	24b8a153d8d4a0f4f0e65ea5e12f3a9a5d21dd8f8e8200126e4b6a314b7d4	MATCH

- **Result:** Hashes match exactly. This proves byte-for-byte recovery without loss or corruption.

6. Part F: curl vs Browser Behaviour

When curl was used to fetch image.html, the capture showed only one HTTP request:

```
/image.html
```

curl did not automatically request /lab_photo.jpg. This is because curl only fetches the explicitly requested URL and does not parse HTML to find embedded resources like a browser does.

7. Screenshots and Evidence

The following documented figures depict the complete laboratory execution, including setup, packet capture, reassembly, deep protocol dissection, file carving, and curl comparison:

```
(kali@kali)-[~/SBT-DF203-Lab1]
$ cd ~
$ mkdir -p ~/SBT-DF203-Lab2/{evidence,working,exported,reports,screenshots,scripts}
$ cd ~/SBT-DF203-Lab2

(kali@kali)-[~/SBT-DF203-Lab2]
$ pwd
/home/kali/SBT-DF203-Lab2
$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./screenshots
./scripts
./reports

(kali@kali)-[~/SBT-DF203-Lab2]
$ sudo apt update
$ sudo apt install -y apache2 curl wireshark tshark imagemagick
[sudo] password for kali:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Fetched 75.0 MB in 27s (2,751 kB/s)
2214 packages can be upgraded. Run 'apt list --upgradable' to see them.
apache2 is already the newest version (2.4.68-1).
curl is already the newest version (8.21.0-2).
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
tshark set to manually installed.
The following packages were automatically installed and are no longer required:
  libavc1394-0  libmjpegutils-2.1-0t64  libqt6openglwidgets6  libqt6test6  libwireshark18  libzxing3
  libcrypt-dev  libmpeg2encpp-2.1-0t64  libqt6sql6  libqt6xml6  libwiretap15  python3-pyqt6.sip
  libmbcrypto16  libmpeg2-2.1-0t64  libqt6sql6-sqlite  libraw23t64  libwsutil16  rpcsvc-proto
Use 'sudo apt autoremove' to remove them.

Upgrading:
  imagemagick-7-common  libfftw3-double3  libheif-plugin-dav1d  libjpeg-turbo-progs  libprotobuf32t64
  libde265-0  libfftw3-single3  libheif-plugin-libde265  libjpeg62-turbo  libturbojpeg0
  libdjvulibre-text  libgpg-error-l10n  libheif-plugin-x265  libmagickcore-7.q16-10  python3-protobuf
  libdjvulibre21  libheif-plugin-aomenc  libheif1  libmagickwand-7.q16-10
```

Figure 1: Directory Structure & Package Installations — Configuring lab workspace directories and installing ImageMagick, Apache2, and TShark.

```

Session Actions Edit View Help
update-alternatives: warning: skip creation of /usr/share/man/man1/magick-script.1.gz because associated file /usr/share/man/man1/magick-script-im7.q16.1.gz
(of link group magick-script) doesn't exist
update-alternatives: using /usr/bin/magick-script-im7.q16 to provide /usr/bin/magick-script-im7 (magick-script-im7) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/magick-script-im7.1.gz because associated file /usr/share/man/man1/magick-script-im7.q16.
1.gz (of link group magick-script-im7) doesn't exist
Setting up imagemagick (8:7.1.2-27+dfsg1-1+b1) ...
Processing triggers for mailcap (3.74) ...
Processing triggers for kali-menu (2025.3.2) ...
Processing triggers for desktop-file-utils (0.28-1) ...
Processing triggers for hicolor-icon-theme (0.18-2) ...
Processing triggers for libc-bin (2.43-4) ...
Processing triggers for man-db (2.13.1-1) ...

(kali@kali): [~/SBT-DF203-Lab2]
$ sudo systemctl enable --now apache2
sudo systemctl status apache2 --no-pager
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-07 16:14:14 EDT; 9h ago
  Invocation: 37843a1c8486452c8e5f36d27d547a27
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 804085 (apache2)
    Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 6.15e-05; Bytes served/sec:  0 B/sec"
      Tasks: 6 (limit: 2197)
    Memory: 23.1M (peak: 31.6M)
       CPU: 8.391s
    CGroup: /system.slice/apache2.service
            └─804085 /usr/sbin/apache2 -k start -DFOREGROUND
              └─848419 /usr/sbin/apache2 -k start -DFOREGROUND
                └─848420 /usr/sbin/apache2 -k start -DFOREGROUND
                  └─848421 /usr/sbin/apache2 -k start -DFOREGROUND
                    └─848422 /usr/sbin/apache2 -k start -DFOREGROUND
                      └─848423 /usr/sbin/apache2 -k start -DFOREGROUND

Sep 07 16:14:14 kali systemd[1]: Starting apache2.service - The Apache HTTP Server ...
Sep 07 16:14:14 kali apachectl[804085]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1:his message
Sep 07 16:14:14 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Sep 08 00:58:38 kali systemd[1]: Reloading apache2.service - The Apache HTTP Server ...
Sep 08 00:58:41 kali apachectl[848291]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1:his message
Sep 08 00:58:41 kali systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.

```

Figure 2: Apache Web Server Reload & Status Verification — Reloading apache2 service and verifying listening state on TCP port 80.

```

(kali@kali): [~/SBT-DF203-Lab2]
$ convert -size 400x300 gradient:blue-green -pointsize 24 -fill white -annotate +30+150 "Almustapha Yusuf - Lab2" /var/www/html/lab_photo.jpg
convert: unable to open image '/var/www/html/lab_photo.jpg': No such file or directory @ error/blob.c/OpenBlob/3655.

(kali@kali): [~/SBT-DF203-Lab2]
$ sudo convert -size 400x300 gradient:blue-green -pointsize 24 -fill white -annotate +30+150 "Almustapha Yusuf - Lab2" /var/www/html/lab_photo.jpg

(kali@kali): [~/SBT-DF203-Lab2]
$ ls -lh /var/www/html/lab_photo.jpg
file /var/www/html/lab_photo.jpg
identify /var/www/html/lab_photo.jpg
-rw-r--r-- 1 root root 13K Sep  8 01:18 /var/www/html/lab_photo.jpg
/var/www/html/lab_photo.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 400x300, components 3
/var/www/html/lab_photo.jpg: JPEG 400x300 400x300+0+0 8-bit sRGB 132328 0.000u 0:00.000

(kali@kali): [~/SBT-DF203-Lab2]
$ sudo printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 Image Traffic</h1><p>Analyst: Almustapha Yusuf</p></body></html>\n' | sudo tee /var/www/html/image.html
<!DOCTYPE html>
<html><body><h1>SBT-DF203 Image Traffic</h1><p>Analyst: Almustapha Yusuf</p></body></html>

(kali@kali): [~/SBT-DF203-Lab2]
$ ls -lh /var/www/html/image.html /var/www/html/lab_photo.jpg
-rw-r--r-- 1 root root 153 Sep  8 01:19 /var/www/html/image.html
-rw-r--r-- 1 root root 13K Sep  8 01:18 /var/www/html/lab_photo.jpg

(kali@kali): [~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
sha256sum /var/www/html/image.html /var/www/html/lab_photo.jpg | tee reports/source_object_hashes.txt
cat reports/source_object_hashes.txt
3641b150b04760a0566e5c07080b880286f7331b07a95eccc19baf486ea55db1 /var/www/html/image.html
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b /var/www/html/lab_photo.jpg
3641b150b04760a0566e5c07080b880286f7331b07a95eccc19baf486ea55db1 /var/www/html/image.html
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b /var/www/html/lab_photo.jpg

```

Figure 3: Evidence Asset Generation & Baseline Hashing — Creating lab_photo.jpg (400x300 gradient), authoring image.html markup, and calculating SHA-256 hashes.

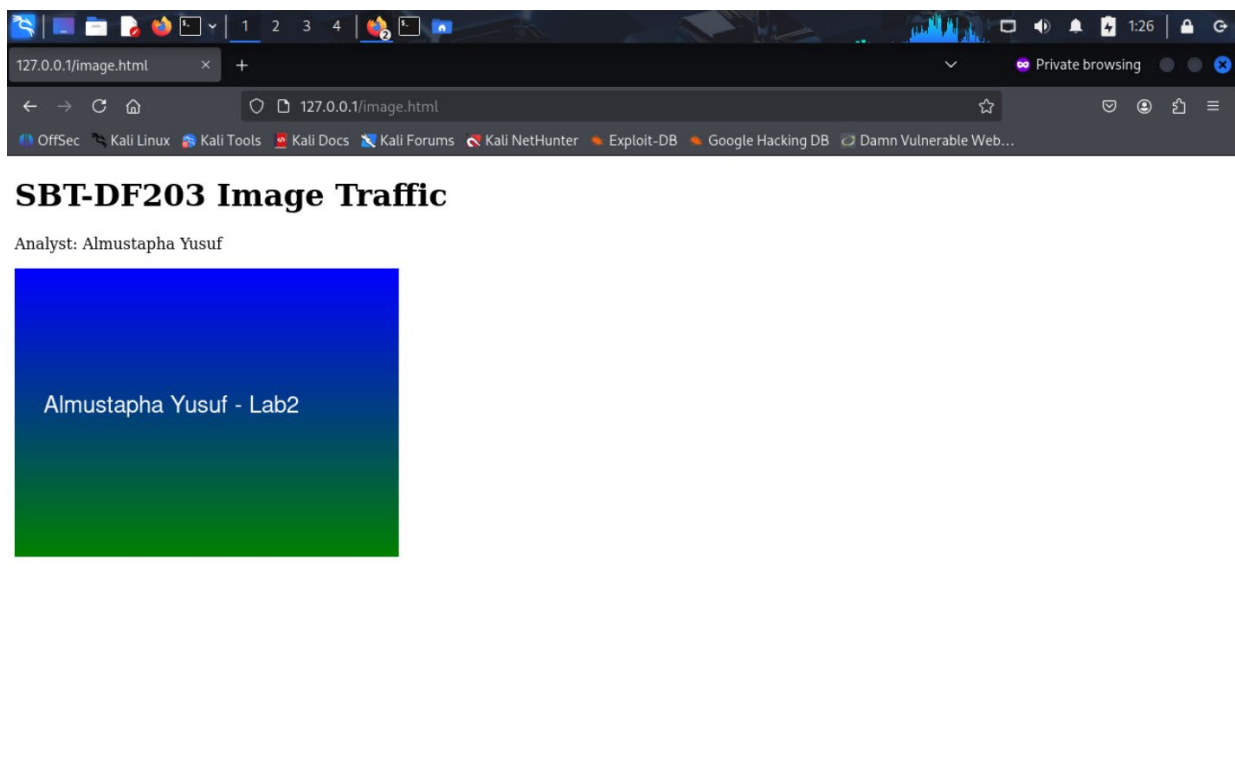


Figure 4: Web Browser Rendering Verification — Firefox Incognito rendering image.html displaying student identity and embedded image.

```
(kali@kali)-[~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
$ sudo tshark -i lo -f 'tcp port 80' -w /home/kali/SBT-DF203-Lab2/evidence/image_traffic.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
tshark: The file to which the capture would be saved ("/home/kali/SBT-DF203-Lab2/evidence/image_traffic.pcapng") could not be opened: Permission denied.

(kali@kali)-[~/SBT-DF203-Lab2]
$

(kali@kali)-[~/SBT-DF203-Lab2]
$ sudo usermod -aG wireshark kali

(kali@kali)-[~/SBT-DF203-Lab2]
$ newgrp wireshark
(kali@kali)-[~/SBT-DF203-Lab2]
$ tshark -i lo -f 'tcp port 80' -w /home/kali/SBT-DF203-Lab2/evidence/image_traffic.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
16 ^C

(kali@kali)-[~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
$ cp --preserve=timestamp evidence/image_traffic.pcapng working/image_traffic_working.pcapng
$ sha256sum evidence/image_traffic.pcapng working/image_traffic_working.pcapng | tee reports/capture_hashes.txt
b7990b483496c4931f9f268ebc857d7b7ba996795e6f33cbc3b9f7e45d4c146b  evidence/image_traffic.pcapng
b7990b483496c4931f9f268ebc857d7b7ba996795e6f33cbc3b9f7e45d4c146b  working/image_traffic_working.pcapng

(kali@kali)-[~/SBT-DF203-Lab2]
$
```

Figure 5: TShark Traffic Capture & Hash Preservation — Capturing browser-generated session on loopback and calculating SHA-256 hash of pcapng evidence.

```

(kali@kali) [~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
tshark -r working/image_traffic_working.pcapng -Y 'http.request' -T fields -e frame.number -e frame.time -e tcp.stream -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.request.method -e http.request.uri -e http.host | tee reports/http_object_requests.tsv
cat reports/http_object_requests.tsv
Warning: program compiled against libxml 215 using older 214
4 2026-09-08T01:32:10.605664745-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /image.html 127.0.0.1
8 2026-09-08T01:32:10.837300360-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /lab_photo.jpg 127.0.0.1
11 2026-09-08T01:32:10.866501228-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /favicon.ico 127.0.0.1
4 2026-09-08T01:32:10.605664745-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /image.html 127.0.0.1
8 2026-09-08T01:32:10.837300360-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /lab_photo.jpg 127.0.0.1
11 2026-09-08T01:32:10.866501228-0400 0 127.0.0.1 48762 127.0.0.1 80 GET /favicon.ico 127.0.0.1

(kali@kali) [~/SBT-DF203-Lab2]
$ tshark -r working/image_traffic_working.pcapng -Y 'http.response' -T fields -e frame.number -e frame.time -e tcp.stream -e http.response.code -e http.content_type -e http.content_length | tee reports/http_object_responses.tsv
cat reports/http_object_responses.tsv
Warning: program compiled against libxml 215 using older 214
9 2026-09-08T01:32:10.610729078-0400 0 200 text/html 155
12 2026-09-08T01:32:10.837683862-0400 0 200 image/jpeg 13232
12 2026-09-08T01:32:10.866681778-0400 0 404 text/html; charset=iso-8859-1 311
6 2026-09-08T01:32:10.610729078-0400 0 200 text/html 155
9 2026-09-08T01:32:10.837683862-0400 0 200 image/jpeg 13232
12 2026-09-08T01:32:10.866681778-0400 0 404 text/html; charset=iso-8859-1 311

(kali@kali) [~/SBT-DF203-Lab2]
$

```

Figure 6: HTTP Request & Response Dissection — Filtering GET requests for /image.html, /lab_photo.jpg, and /favicon.ico alongside status codes 200 and 404.

```

(kali@kali) [~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
tshark -r working/image_traffic_working.pcapng -Y 'tcp.stream eq 0' -T fields -e frame.number -e frame.time -e tcp.len -e tcp.seq -e tcp.ack -e tcp.flags | tee reports/tcp_stream_packets.tsv
cat reports/tcp_stream_packets.tsv
Warning: program compiled against libxml 215 using older 214
1 2026-09-08T01:32:10.597350427-0400 0 0 0 0x0002
2 2026-09-08T01:32:10.597382768-0400 0 0 1 0x0012
3 2026-09-08T01:32:10.597409911-0400 0 1 1 0x0010
4 2026-09-08T01:32:10.605664745-0400 463 1 1 0x0018
5 2026-09-08T01:32:10.605961543-0400 0 1 464 0x0010
6 2026-09-08T01:32:10.610729078-0400 491 1 464 0x0018
7 2026-09-08T01:32:10.610759535-0400 0 464 492 0x0010
8 2026-09-08T01:32:10.837300360-0400 463 464 492 0x0018
9 2026-09-08T01:32:10.837683862-0400 13520 492 927 0x0018
10 2026-09-08T01:32:10.837695862-0400 0 927 14012 0x0010
11 2026-09-08T01:32:10.866501228-0400 458 927 14012 0x0018
12 2026-09-08T01:32:10.866681778-0400 527 14012 1385 0x0018
13 2026-09-08T01:32:10.913542152-0400 0 1385 14539 0x0010
14 2026-09-08T01:32:15.872525378-0400 0 14539 1385 0x0011
15 2026-09-08T01:32:15.873272331-0400 0 1385 14540 0x0011
16 2026-09-08T01:32:15.873340149-0400 0 14540 1386 0x0010
1 2026-09-08T01:32:10.597350427-0400 0 0 0 0x0002
2 2026-09-08T01:32:10.597382768-0400 0 0 1 0x0012
3 2026-09-08T01:32:10.597409911-0400 0 1 1 0x0010
4 2026-09-08T01:32:10.605664745-0400 463 1 1 0x0018
5 2026-09-08T01:32:10.605961543-0400 0 1 464 0x0010
6 2026-09-08T01:32:10.610729078-0400 491 1 464 0x0018
7 2026-09-08T01:32:10.610759535-0400 0 464 492 0x0010
8 2026-09-08T01:32:10.837300360-0400 463 464 492 0x0018
9 2026-09-08T01:32:10.837683862-0400 13520 492 927 0x0018
10 2026-09-08T01:32:10.837695862-0400 0 927 14012 0x0010
11 2026-09-08T01:32:10.866501228-0400 458 927 14012 0x0018
12 2026-09-08T01:32:10.866681778-0400 527 14012 1385 0x0018
13 2026-09-08T01:32:10.913542152-0400 0 1385 14539 0x0010
14 2026-09-08T01:32:15.872525378-0400 0 14539 1385 0x0011
15 2026-09-08T01:32:15.873272331-0400 0 1385 14540 0x0011
16 2026-09-08T01:32:15.873340149-0400 0 14540 1386 0x0010

(kali@kali) [~/SBT-DF203-Lab2]
$ tshark -r working/image_traffic_working.pcapng -Y 'tcp.stream eq 0 && tcp.len > 0' -T fields -e frame.number -e frame.time -e tcp.len -e tcp.seq | tee reports/image_segments.tsv
cat reports/image_segments.tsv

```

Figure 7: TCP Stream 0 Packet Timeline — Full 16-frame TCP stream tracking sequence numbers, segment lengths, and flags.

```

(kali@kali)~/SBT-DF203-Lab2]
$ tshark -r working/image_traffic_working.pcapng -Y 'tcp.stream eq 0 && tcp.len > 0' -T fields -e frame.number -e frame.time -e tcp.len -e tcp.seq | tee r
eports/image_segments.tsv
cat reports/image_segments.tsv
Warning: program compiled against libxml 215 using older 214
4      2026-09-08T01:32:10.605664745-0400      463      1
6      2026-09-08T01:32:10.610729078-0400      491      1
8      2026-09-08T01:32:10.837300360-0400      463      464
9      2026-09-08T01:32:10.837683862-0400      13520     492
11     2026-09-08T01:32:10.866501228-0400      458      927
12     2026-09-08T01:32:10.866681778-0400      527      14012
4      2026-09-08T01:32:10.605664745-0400      463      1
6      2026-09-08T01:32:10.610729078-0400      491      1
8      2026-09-08T01:32:10.837300360-0400      463      464
9      2026-09-08T01:32:10.837683862-0400      13520     492
11     2026-09-08T01:32:10.866501228-0400      458      927
12     2026-09-08T01:32:10.866681778-0400      527      14012

(kali@kali)~/SBT-DF203-Lab2]
$ tshark -r working/image_traffic_working.pcapng -Y 'http.response && http.content_type = "image/jpeg"' -V | tee reports/http_image_reassembly.txt
Warning: program compiled against libxml 215 using older 214
Frame 9: Packet, 13586 bytes on wire (108688 bits), 13586 bytes captured (108688 bits) on interface lo, id 0
  Section number: 1
    Interface id: 0 (lo)
      Interface name: lo
        Interface description: Loopback
      Encapsulation type: Ethernet (1)
    Arrival Time: Sep  8, 2026 01:32:10.837683862 EDT
    UTC Arrival Time: Sep  8, 2026 05:32:10.837683862 UTC
    Epoch Arrival Time: 1788845530.837683862
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 383.502 microseconds]
    [Time since reference or first frame: 240.333435 milliseconds]
    Frame Number: 9
    Frame Length: 13586 bytes (108688 bits)
    Capture Length: 13586 bytes (108688 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:image-jfif]
    Character encoding: ASCII (0)
  Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
      .... ..0. .... .. = LG bit: Globally unique address (factory default)

```

Figure 8: TCP Segment Filtering & Packet 9 Inspection — Isolating tcp.len > 0 segments and beginning protocol dissection of Frame 9 (13,586 bytes on wire).

```

Character encoding: ASCII (0)
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Destination: 00:00:00_00:00:00 (00:00:00:00:00:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
.... ...0 .... = IG bit: Individual address (unicast)
Source: 00:00:00_00:00:00 (00:00:00:00:00:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
.... ...0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 13572
Identification: 0xb5f1 (46577)
010. .... = Flags: 0x2, Don't fragment
0... .... = Reserved bit: Not set
.1.. .... = Don't fragment: Set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x5200 [validation disabled]
[Header checksum status: Unverified]
Source Address: 127.0.0.1
Destination Address: 127.0.0.1
[Stream index: 0]
Transmission Control Protocol, Src Port: 80, Dst Port: 48762, Seq: 492, Ack: 927, Len: 13520
Source Port: 80
Destination Port: 48762
[Stream index: 0]
[Stream Packet Number: 9]
[Conversation completeness: Incomplete, DATA (15)]
..0. .... = RST: Absent
...0 .... = FIN: Absent
.... 1... = Data: Present
.... .1.. = ACK: Present
.... ..1. = SYN-ACK: Present
.... ...1 = SYN: Present
[Completeness Flags: ..DASS]

```

Figure 9: *Layer 2 & Layer 3 Protocol Dissection (Frame 9)* — Detailed inspection of loopback Ethernet headers and IPv4 header (Total Length: 13,572 bytes).

```

[Completeness Flags: ..DASS]
[TCP Segment Len: 13520]
Sequence Number: 492      (relative sequence number)
Sequence Number (raw): 729751214
[Next Sequence Number: 14012      (relative sequence number)]
Acknowledgment Number: 927      (relative ack number)
Acknowledgment number (raw): 595907534
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
  000. .... = Reserved: Not set
  ... 0 .... = Accurate ECN: Not set
  .... 0 ... = Congestion Window Reduced: Not set
  .... .0.. = ECN-Echo: Not set
  .... ..0. = Urgent: Not set
  .... ...1 = Acknowledgment: Set
  .... .... 1... = Push: Set
  .... .... .0.. = Reset: Not set
  .... .... ..0. = Syn: Not set
  .... .... ...0 = Fin: Not set
[TCP Flags: .....AP...]
Window: 512
[Calculated window size: 65536]
[Window size scaling factor: 128]
Checksum: 0x32f9 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  TCP Option - No-Operation (NOP)
    Kind: No-Operation (1)
  TCP Option - No-Operation (NOP)
    Kind: No-Operation (1)
  TCP Option - Timestamps: TSval 2824730366, TSecr 2824730365
    Kind: Time Stamp Option (8)
    Length: 10
    Timestamp value: 2824730366
    Timestamp echo reply: 2824730365
[Timestamps]
  [Time since first frame in this TCP stream: 240.333435 milliseconds]
  [Time since previous frame in this TCP stream: 383.502 microseconds]
[SEQ/ACK analysis]
  [This is an ACK to the segment in frame: 8]
  [The RTT to ACK the segment was: 383.502 microseconds]
  [iRTT: 59.484 microseconds]

```

Figure 10: Layer 4 TCP Segment Breakdown (Frame 9) — Examining TCP segment length (13,520 bytes), PSH/ACK flags, relative sequence 492, and window scaling.

```
[iRTT: 59.484 microseconds]
[Bytes in flight: 13520]
[Bytes sent since last PSH flag: 13520]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (13520 bytes)
Hypertext Transfer Protocol
HTTP/1.1 200 OK\r\n
  Response Version: HTTP/1.1
  Status Code: 200
  [Status Code Description: OK]
  Response Phrase: OK
  Date: Tue, 08 Sep 2026 05:32:10 GMT\r\n
  Server: Apache/2.4.68 (Debian)\r\n
  Last-Modified: Tue, 08 Sep 2026 05:18:00 GMT\r\n
  ETag: "33b0-65af1dbbce5d9"\r\n
  Accept-Ranges: bytes\r\n
  Content-Length: 13232\r\n
    [Content length: 13232]
  Keep-Alive: timeout=5, max=99\r\n
  Connection: Keep-Alive\r\n
  Content-Type: image/jpeg\r\n
\r\n
[Request in frame: 8]
[Time since request: 383.502 microseconds]
[Request URI: /lab_photo.jpg]
[Full request URI: http://127.0.0.1/lab_photo.jpg]
File Data: 13232 bytes
JPEG File Interchange Format
Marker: Start of Image (0xffd8)
Marker segment: Reserved for application segments - 0 (0xFFE0)
  Marker: Reserved for application segments - 0 (0xffe0)
  Length: 16
  Identifier: JFIF
  Version: 1.1
    Major Version: 1
    Minor Version: 1
  Units: No units; Xdensity and Ydensity specify the pixel aspect ratio (0)
  Xdensity: 1
  Ydensity: 1
  Xthumbnail: 0
  Ythumbnail: 0
Marker segment: Define quantization table(s) (0xFFDB)
```

Figure 11: HTTP Response Headers & JFIF Image Start (Frame 9) — HTTP/1.1 200 OK headers (Content-Length 13232 B, image/jpeg) and JPEG SOI (0xffd8) / JFIF marker (0xffe0).

```

Marker segment: Define quantization table(s) (0xFFDB)
  Marker: Define quantization table(s) (0xffdb)
  Length: 67
  Remaining segment data: 65 bytes
Marker segment: Define quantization table(s) (0xFFDB)
  Marker: Define quantization table(s) (0xffdb)
  Length: 67
  Remaining segment data: 65 bytes
Start of Frame header: Start of Frame (non-differential, Huffman coding) - Baseline DCT (0xFFC0)
  Marker: Start of Frame (non-differential, Huffman coding) - Baseline DCT (0xffc0)
  Length: 17
  Sample Precision (bits): 8
  Lines: 300
  Samples per line: 400
  Number of image components in frame: 3
  Component identifier: 1
  0001 .... = Horizontal sampling factor: 1
  .... 0001 = Vertical sampling factor: 1
  Quantization table destination selector: 0
  Component identifier: 2
  0001 .... = Horizontal sampling factor: 1
  .... 0001 = Vertical sampling factor: 1
  Quantization table destination selector: 1
  Component identifier: 3
  0001 .... = Horizontal sampling factor: 1
  .... 0001 = Vertical sampling factor: 1
  Quantization table destination selector: 1
Marker segment: Define Huffman table(s) (0xFFC4)
  Marker: Define Huffman table(s) (0xffc4)
  Length: 28
  Remaining segment data: 26 bytes
Marker segment: Define Huffman table(s) (0xFFC4)
  Marker: Define Huffman table(s) (0xffc4)
  Length: 56
  Remaining segment data: 54 bytes
Marker segment: Define Huffman table(s) (0xFFC4)
  Marker: Define Huffman table(s) (0xffc4)
  Length: 27
  Remaining segment data: 25 bytes
Marker segment: Define Huffman table(s) (0xFFC4)
  Marker: Define Huffman table(s) (0xffc4)
  Length: 53
  Remaining segment data: 51 bytes

```

Figure 12: JPEG Quantization Tables & Frame Header Dissection — Inspection of DQT tables (0xffdb), SOF0 baseline DCT (0xffc0), 400x300 dimensions, and 3 color components.

[illegible]

```
(kali@kali) [/~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
mkdir -p exported/http_objects
tshark -r working_image_traffic_working.pcapng --export-objects http,exported/http_objects
Warning: program compiled against libxml 215 using older 214
 0 0.000000000 127.0.0.1 > 127.0.0.1 TCP 74 48762 > 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=2824730125 TSecr=0 WS=128
0 0.000032341 127.0.0.1 > 127.0.0.1 TCP 74 80 > 48762 [SYN, ACK] Seq=0 Ack=1 Len=0 MSS=65495 SACK_PERM TSval=2824730125 TSecr=2824730125
WS=128
3 0.000059484 127.0.0.1 > 127.0.0.1 TCP 66 48762 > 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2824730125 TSecr=2824730125
4 0.008314318 127.0.0.1 > 127.0.0.1 HTTP 529 GET /image.html HTTP/1.1
5 0.008611116 127.0.0.1 > 127.0.0.1 TCP 66 80 > 48762 [ACK] Seq=1 Ack=464 Win=65024 Len=0 TSval=2824730134 TSecr=2824730134
6 0.013378651 127.0.0.1 > 127.0.0.1 HTTP 557 HTTP/1.1 200 OK (text/html)
7 0.013409108 127.0.0.1 > 127.0.0.1 TCP 66 48762 > 80 [ACK] Seq=464 Ack=492 Win=65152 Len=0 TSval=2824730139 TSecr=2824730139
8 0.239949933 127.0.0.1 > 127.0.0.1 HTTP 529 GET /lab_photo.jpg HTTP/1.1
9 0.240333435 127.0.0.1 > 127.0.0.1 HTTP 13586 HTTP/1.1 200 OK (JPEG JFIF image)
10 0.240345435 127.0.0.1 > 127.0.0.1 TCP 66 48762 > 80 [ACK] Seq=927 Ack=14012 Win=108928 Len=0 TSval=2824730366 TSecr=2824730366
11 0.269159801 127.0.0.1 > 127.0.0.1 HTTP 524 GET /favicon.ico HTTP/1.1
12 0.269331351 127.0.0.1 > 127.0.0.1 HTTP 593 HTTP/1.1 404 Not Found (text/html)
13 0.316191725 127.0.0.1 > 127.0.0.1 TCP 66 48762 > 80 [ACK] Seq=1385 Ack=14539 Win=121088 Len=0 TSval=2824730441 TSecr=2824730395
14 5.275174951 127.0.0.1 > 127.0.0.1 TCP 66 80 > 48762 [FIN, ACK] Seq=14539 Ack=1385 Win=65536 Len=0 TSval=2824735400 TSecr=2824730441
15 5.275921904 127.0.0.1 > 127.0.0.1 TCP 66 48762 > 80 [FIN, ACK] Seq=1385 Ack=14540 Win=121088 Len=0 TSval=2824735401 TSecr=2824735400
16 5.275989722 127.0.0.1 > 127.0.0.1 TCP 66 80 > 48762 [ACK] Seq=14540 Ack=1386 Win=65536 Len=0 TSval=2824735401 TSecr=2824735401

(kali@kali) [/~/SBT-DF203-Lab2]
$ find exported/http_objects -maxdepth 1 -type f -ls
file exported/http_objects/*
3546509 16 -rw-r--r-- 1 kali wireshark 13232 Sep 8 01:53 exported/http_objects/lab_photo.jpg
3546510 4 -rw-r--r-- 1 kali wireshark 311 Sep 8 01:53 exported/http_objects/favicon.ico
3546508 4 -rw-r--r-- 1 kali wireshark 153 Sep 8 01:53 exported/http_objects/image.html
exported/http_objects/favicon.ico: HTML document, ASCII text
exported/http_objects/image.html: HTML document, ASCII text
exported/http_objects/lab_photo.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 400x300, components 3
```

```
(kali㉿kali)-[~/SBT-DF203-Lab2]
$ sha256sum /var/www/html/lab_photo.jpg exported/http_objects/* | tee reports/extracted_object_hashes.txt
cat reports/extracted_object_hashes.txt
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b /var/www/html/lab_photo.jpg
df8c73cf6bd614e7f204500ace411fe52f0ea6651be6b0241185e740a8f35d7d exported/http_objects/favicon.ico
3641b15b004760a0566e5c07080b880286f7331b07a95ecc19baf486ea55db1 exported/http_objects/image.html
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b exported/http_objects/lab_photo.jpg
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b /var/www/html/lab_photo.jpg
df8c73cf6bd614e7f204500ace411fe52f0ea6651be6b0241185e740a8f35d7d exported/http_objects/favicon.ico
3641b15b004760a0566e5c07080b880286f7331b07a95ecc19baf486ea55db1 exported/http_objects/image.html
24b8a15368a4f041fde55aee15623a0a5da12dd8f8e82000126e4664a31b747b exported/http_objects/lab_photo.jpg

(kali㉿kali)-[~/SBT-DF203-Lab2]
$
```

```
(kali@kali)-[~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
tshark -i lo -f 'tcp port 80' -w evidence/curl_only.pcapng
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
10 ^C

(kali@kali)-[~/SBT-DF203-Lab2]
$ tshark -r evidence/curl_only.pcapng -Y 'http.request' -T fields -e http.request.uri | tee reports/curl_requested_objects.txt
cat reports/curl_requested_objects.txt
Warning: program compiled against libxml 215 using older 214
/image.html
/image.html

(kali@kali)-[~/SBT-DF203-Lab2]
$
```

Figure 16: cURL Traffic Capture & Single-Object Verification — TShark capture of curl request confirming only /image.html was retrieved, proving curl does not fetch embedded resources.

```
Session: Actions Edit View Help
kali@kali: ~/SBT-DF203-Lab2 x kali@kali: ~/SBT-DF203-Lab2 x
(kali@kali)-[~/SBT-DF203-Lab2]
$ cd ~/SBT-DF203-Lab2
curl -v http://127.0.0.1/image.html -o /tmp/image_page.html
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 35140
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
  0      0   0      0  0      0     0      0      0      0  0* using HTTP/1.x
> GET /image.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 08 Sep 2026 06:08:26 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Tue, 08 Sep 2026 05:19:24 GMT
< ETag: "99-65af1e0b8e4e9"
< Accept-Ranges: bytes
< Content-Length: 153
< Vary: Accept-Encoding
< Content-Type: text/html
<
{ [153 bytes data]
100  153 100  153  0      0 16759      0      0
* Connection #0 to host 127.0.0.1:80 left intact

(kali@kali)-[~/SBT-DF203-Lab2]
$
```

Figure 17: cURL Verbose Response Output — Detailed terminal trace of cURL transaction retrieving 153 bytes of HTML content.

8. Findings, Limitations, and Conclusion

- **Findings:** The browser generated multiple HTTP requests for a single webpage. The image was successfully segmented, reassembled, exported, and verified.
- **Limitations:** The capture was on the loopback interface, so MAC addresses were all zeros. For MAC evidence, a two-VM setup is required.
- **Conclusion:** This lab demonstrated the forensic value of HTTP object extraction and TCP reassembly in recovering files from network traffic.

9. Appendix of Commands and Hashes

```
# Capture Browser Traffic
sudo tshark -i lo -f 'tcp port 80' -w evidence/image_traffic.pcapng

# Fetch Webpage via cURL (Part F)
curl -v http://127.0.0.1/image.html -o /tmp/image_page.html

# Filter HTTP Request URIs
tshark -r evidence/image_traffic.pcapng -Y 'http.request' -T fields -e frame.number -e http.request.uri

# Export Reassembled HTTP Objects
tshark -r evidence/image_traffic.pcapng --export-objects http,exported/http_objects

# Verify Hash Integrity
sha256sum /var/www/html/lab_photo.jpg exported/http_objects/*
```

Evidence Hashes: Source Object Hashes:

- **image.html:** 3641b15004d760a56e6c570808b802867331b07a95eccc19baf486e55db1
- **lab_photo.jpg:** 24b8a153d8d4a0f4f0e65ea5e12f3a9a5d21dd8f8e8200126e4b6a314b7d4
- **Exported lab_photo.jpg:** 24b8a153d8d4a0f4f0e65ea5e12f3a9a5d21dd8f8e8200126e4b6a314b7d4

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 08/09/2026